

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element G Initial Template

Element G: Construction of a Testable Prototype

Final Prototype Iteration Explanation (G1)

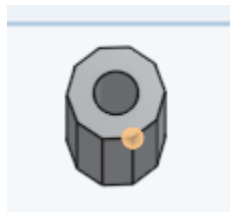
3D Print Design

Date: 4/16

Photos:



Idea #1



idea #2

	Details	Tools and Manufacturing Methods used
Step 1	Go onto on shape and design the shape of how thick, how big and how wide you want the attachable piece	computer and onshape account
Step 2	Upload to a 3d printer and print out two of them	3d printer

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Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

- Button design for strings or pipe cleaners to be strung through to adjust to each size



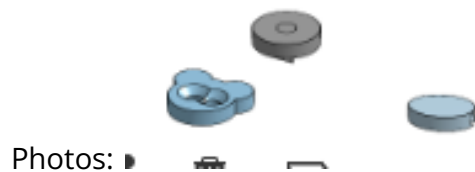
Was able to get this on the string but took a very long time and a lot of poking with a sewing needle. Easy to get off and does not stay on. Very hard to get onto the string



once the pipecleaner and string was looped through, the device stayed on nicely. But due to the tightness of the knot and the pipecleaner, the string broke. This could have been because of the sharp edges of the hole as well.

Design #2 with iterations

Date: 4/16/24



Photos: 

3d print design

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	Details	Tools and Manufacturing Methods used
Step 1	Use cad to draw out the structure	Onshape
Step 2	Uploaded to the computer	Computer 3d print software
Step 3	Upload to 3d printer	3d printer
Step 4	Print out the device	3d printer

Notes of the progression:

- the first was too small, was not able to get the string though

Second one was made larger

Made it so the wearer could add drawings, stickers or other decals.

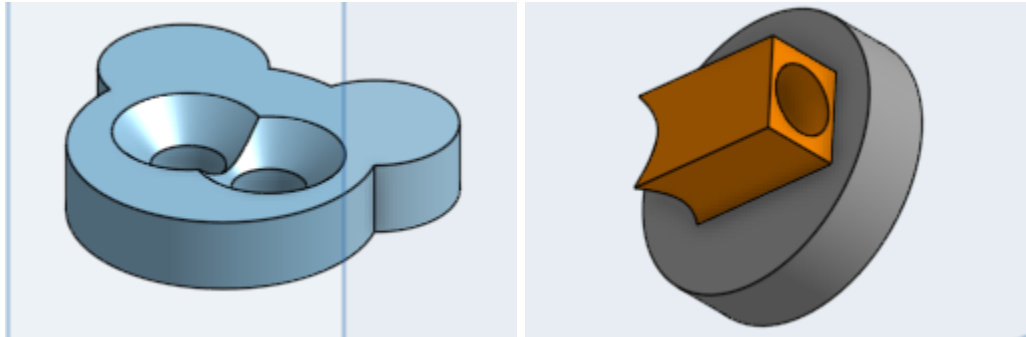
Material name	Description	Where material was purchased	Cost per unit	# units	Total cost per item
3d print material	Rolls of plastic	From ms. rudnick			

Sketches:

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Construction of the Final Prototype (G2)

Design Requirement	How will this prototype allow you to collect objective, measurable data?
grip tightly onto any drawstring	Use the prototype and put on the hoodie strings of different students and see how it will work
be sturdy	The structured 3d plastic is probably hard enough to test how it deals with drops and crushes
hoodie string does not get lost	We can put the prototype to use and pull the other side of the string to see if the string will be held back
look fashionable	We can ask and survey stakeholders and students to see what they think of the prototype and ask if they would be willing to wear it out
stay on effortlessly	We can time how long it takes for a person to put the prototype onto their hoodie
kid-proof and pet-proof	We can go onto government regulation sights and search how big the item needs to be to qualify under kid proof and pet proof
Not damage the hoodie string	Use the prototype and pull the hoodie string though back and forth to see if there is any fraying of the string

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Testing or Modeling of Attributes (G3, G4) and Addressing Non-Testable Attributes (G4)

Attribute	Description	Describe your prototype's functionality to test this attribute
Dipped in holes	The holes on both prototypes are dipped in so it is easier to put on	Can be timed to see if the dipped in hole is easier to be put on than the one before where there was no dip
Flat surface	Created a flat somewhat smooth surface that can be decorated based on what the stakeholder wants	Can use markers and stickers to see whether it will stick
Grip ability of the device	The holes on the prototype need to be small enough to hold the string in place but also big enough to let the string be put through	Expert feedback required

Addressing Non-Testable Attributes (G4)

For all attributes that you listed "expert review required" in the table above, provide a justification for why that is true. "Field experts" are professionals who spend at least a portion of their time working in a field directly related to your project. These field experts are not permitted to double count as stakeholders.

Attribute	Justification for why this attribute cannot be modeled or tested with your prototype
Grip ability of the device	Although we can test how much it will hold, we cannot



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	test to determine the strength of the grip and how long it will sustain like that until it does not work
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Element G: Constructio n of a Testable Prototype						
	5	4	3	2	1	0
1. Final Prototype Iteration Explanation. The Final Prototype Iteration is _____.	clearly and fully explained	clearly and adequately explained	adequately explained	somewhat explained but with gaps	only minimally explained	is unclear or is missing altogether
2. Construction of the Final Prototype. The Final Prototype Iteration is _____.	constructed with enough detail to assure that objective data on <i>all</i> or nearly all design requirements could be determined	constructed with enough detail to assure that objective data on <i>many or nearly all</i> design requirements could be determined	constructed with enough detail to assure that objective data on <i>some</i> design requirements could be determined	constructed with enough detail to assure that objective data on at least a <i>few</i> design requirements could be determined	<i>not</i> constructed with enough detail to assure that objective data on at least <i>one</i> design requirement could be determined	<i>not applicable</i> to any design requirements and/or functional claims
3. Testing or Modeling of Attributes. _____ attributes (sub-systems) of	<i>All</i>	<i>Most</i>	<i>Some</i>	<i>A Few</i>	<i>No more than one</i>	<i>None of the</i>



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the unique solution that can be tested or modeled mathematically are addressed.						
4.Addressing non-Testable Attributes. justification is provided for those that cannot be tested or modeled mathematically and thus require expert review.	<i>A well supported</i>	<i>A generally supported</i>	<i>An adequately supported</i>	<i>Developing</i>	<i>Insufficient</i>	<i>No</i>